Hybrid Quantum AI Models for Accelerating Multiphysics Simulation- Part 1

For Crystal Growth Simulations and Liquid H_2 Cryogenics

Quasi Team, Collaborators

October 2026

Outline

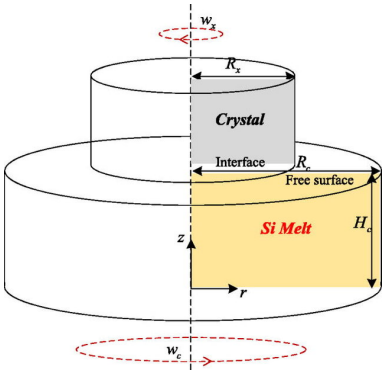
1. Motivation
2. Classical approaches
3. Hybrid quantum–classical model
4. Benchmark and results
5. Outlook

Multiphysics simulation is too slow for design loops

- High-fidelity coupled CFD (flow + heat + rotation + phase change) takes hours to days per run.
- Every new operating point (heater power, rotation rate, fill level) is a new simulation.
- Process control and digital twins need full fields in **milliseconds**.

Goal

A surrogate that keeps the governing physics, generalises across operating conditions, and evaluates in real time.



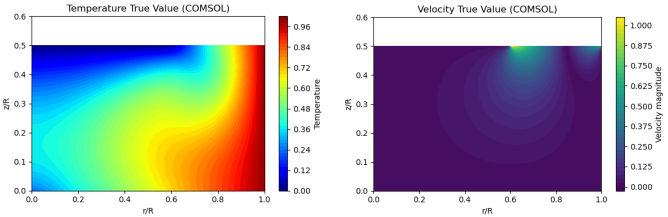
Czochralski (CZ) growth: crystal rotating at ω_s, pulled from a Si melt in a crucible rotating at ω_c.

Accuracy is not the key bottleneck; the design space.

Two industrial use cases

Silicon crystal growth (Czochralski)

Melt convection, swirl and temperature (1685–1750 K) set defect density and oxygen uptake. Five coupled fields per point: u_r, u_z, u_θ, p, T.



COMSOL reference: temperature (left) and velocity magnitude (right). Sharp gradients sit at the free surface and the crystal edge.

LH_2 cryogenic boil-off

- Heat leak into liquid-hydrogen tanks (≈20 K) drives natural convection, stratification and evaporation.
- Turbulent convection with a **moving liquid–vapour interface**.
- Aim: boil-off rate and pressure rise for tank design.

Common thread

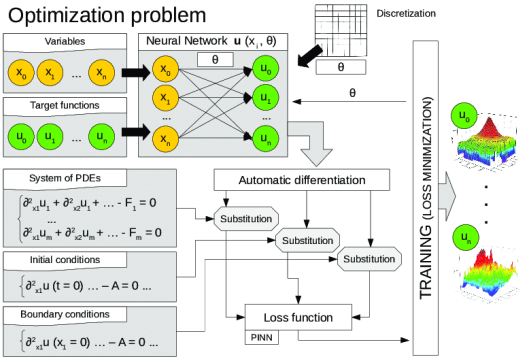
Strongly coupled, non-linear transport of velocity, temperature, species and interfaces.

Key ideas

1. **Problem:** classical PINNs suffer from spectral bias, loss imbalance and badly scaled coupled fields.
2. **Model:** Hybrid-QPINN / QPINO: a classical network with a shallow parameterised quantum circuit (PQC) as a trainable feature map, fed by Fourier features and data re-uploading.
3. **Tools:** PennyLane (training), Qiskit Aer (simulation), Qiskit Runtime (IBM hardware).
4. **Benchmarks:** silicon CZ crystal growth (running) and LH_2 boil-off (defined).
5. **Destination:** real-time digital twins and fast parametric exploration.

Goal: Hybrid quantum surrogates that answer in milliseconds instead of hours.

Physics-informed neural networks (PINNs)



Inputs → network → automatic differentiation → PDE, IC and BC residuals → loss → training.

1. A network $u(x, \theta)$ maps coordinates to the target fields.
2. Automatic differentiation gives exact derivatives, so PDE residuals can be evaluated at any point: no mesh.
3. Residuals of PDEs, initial and boundary conditions are combined into one loss:

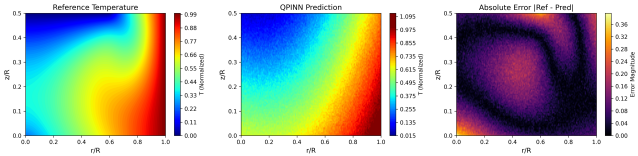
$$\mathcal{L} = \lambda_{\text{PDE}}\mathcal{L}_{\text{PDE}} + \lambda_{\text{IC}}\mathcal{L}_{\text{IC}} + \lambda_{\text{BC}}\mathcal{L}_{\text{BC}}\,(+\,\mathcal{L}_{\text{data}})$$

Limits

One trained PINN = one operating point (Neüral Operator learns familie of fünctions). The weights λ decide what is actually learned.

Bottleneck 1: spectral bias

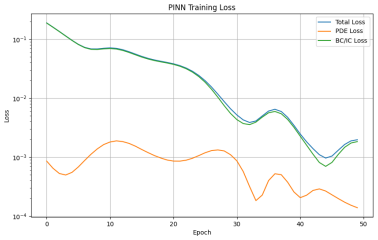
- Neural networks learn low frequencies first; sharp content converges slowly (Rahaman et al., 2019).
- Transport fields are full of sharp structure: thermal boundary layers, swirl near rotating walls, phase interfaces.
- Classical remedy: Fourier feature inputs (Tancik et al., 2020).



Reference temperature, a smooth network prediction and the absolute error, which concentrates on the curved thermal front.

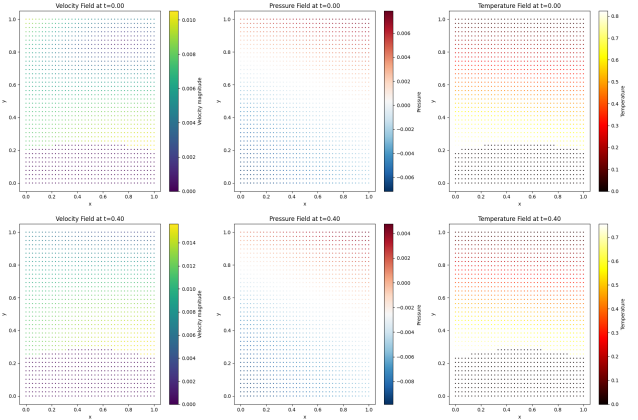
Errors live where the physics is sharpest: the part of the field design decisions depend on.

Classical PINN on simplified crystal growth



Training loss: total, PDE and BC/IC components.

- Total loss follows BC/IC; the PDE term oscillates instead of falling.
- Loss rises again after epoch 45.



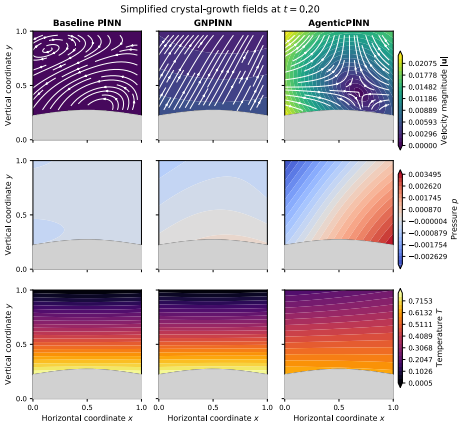
Predicted velocity, pressure and temperature at $t = 0$ (top) and $t = 0.4$ (bottom).

A plain PINN converges to smooth, under-resolved fields.

Bottleneck 2: loss-term imbalance

Trainer (1D heat)	PDE loss	BC/IC loss	Rel. L_2
Baseline	3.96×10^{-3}	1.15×10^{-3}	0.084
GNPINN	9.74×10^{-3}	9.05×10^{-4}	0.054
AgenticPINN	8.22×10^{-7}	2.00×10^{-1}	1.368

- Baseline: fixed weights. GNPINN: gradient-normalised. AgenticPINN: rule-based re-weighting.
- AgenticPINN: PDE residual $\approx$4800$\times$ smaller, BC/IC $\approx$173$\times$ larger, and the solution is wrong.



Simplified CZ fields at $t = 0.2$ for the three trainers (Chapagain et al., 2026).

The smallest PDE loss gave the worst solution: always report per-term and per-field errors.

Bottleneck 3: scale mismatch between coupled fields

- Five outputs u_r, u_z, u_θ, p, T with very different magnitudes.
- In the CZ model the pressure coefficient is

$$p_{\text{coeff}} = \frac{p_{\text{scale}}}{v_{\text{scale}}^2} = 134{,}128,$$

  while viscous terms scale as $1/Re \approx 0.067$.
- Pressure-to-viscous ratio $\approx 2 \times 10^6 : 1$.

Consequence

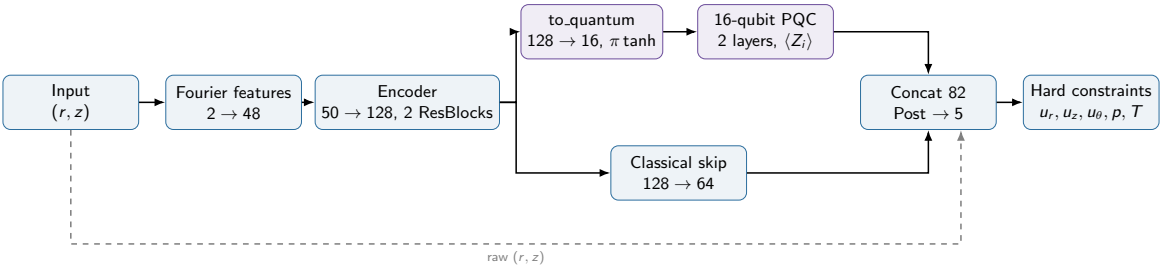
The pressure residual can swamp the data signal during physics-informed training (Result 4).

Design response

Hard constraints, staged physics warm-up, residual balancing, per-field residual scaling.

Non-dimensionalisation is part of the model, not a pre-processing detail.

Architecture overview



Classical backbone

Fourier encoding + residual MLP + post network: 125,925 parameters.

Quantum feature map

16 qubits, 2 layers: **96** trainable angles (0.076% of 126,021).

Skip path

64 classical features bypass the PQC, so the head never depends solely on it.

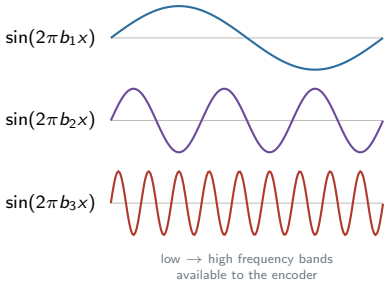
The PQC is a small, expressive feature extractor; the classical network does the heavy lifting.

Ingredient 1: Fourier feature mapping

- Frozen random matrix $B \in \mathbb{R}^{2\times 24}$, $B_{ij} \sim \mathcal{N}(0, 6^2)$:

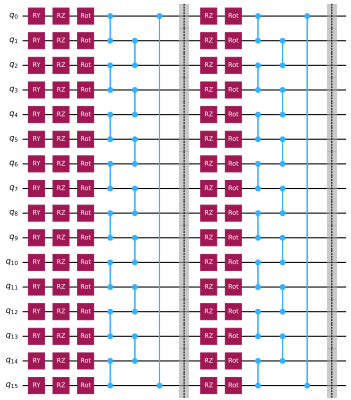
$$\gamma(x) = \big[\sin(2\pi\, xB),\ \cos(2\pi\, xB)\big] \in \mathbb{R}^{48}$$

- Frequencies up to roughly ± 15 cycles per unit length.
- Concatenated with raw (r, z): 50 encoder inputs.
- High-frequency access from the first layer at zero trainable cost.



Fourier features fix the input spectrum; the PQC adds a trainable spectrum deeper in the network.

Ingredient 2: shallow parameterised quantum circuit



Logical circuit: 16 qubits, 2 layers, CZ ring.

1. **Encode:** $R_Y(\theta_i)$, $\theta_i = \pi \tanh(h_i) \in [-\pi, \pi]$.
2. **Re-upload** (each layer ℓ): $R_Z(0.5\,\theta_{(i+\ell) \bmod 16})$.
3. **Train:** $\mathrm{Rot}(\varphi, \vartheta, \omega) = R_Z R_Y R_Z$, 48 angles/layer.
4. **Entangle:** CZ ring q_0–q_1–$\cdots$–q_{15}–q_0.
5. **Read out:** $\langle Z_i \rangle \in [-1, 1] \to$ LayerNorm(16).

R_Y (encode)	16	CZ (entangle)	32
R_Z (re-upload)	32	Total gates	112
Rot (trainable)	32	Logical depth	9

Only 32 two-qubit gates: shallow enough to run reliably on today's hardware.

Ingredient 3: data re-uploading

- A PQC output is a truncated Fourier series of its encoded inputs (Schuld et al., 2021):

$$f_\Phi(\theta) = \sum_{\omega \in \Omega} c_\omega(\Phi)\, e^{i\omega \cdot \theta}$$

- Encoding gates fix the frequencies Ω; trainable gates set the coefficients c_ω.
- Each θ_i enters three times (R_Y once, $R_Z(\frac{1}{2}\theta)$ twice), and the cyclic shift mixes neighbouring latent features in each layer.

Encoding of θ_i	Frequencies
R_Y once	$\{0, \pm 1\}$
+ 2 re-uploads	$\{0, \pm\frac{1}{2}, \ldots, \pm 2\}$
Count	$3 \rightarrow \mathbf{9}$

Re-uploading widens the spectrum the network can represent: the content spectral bias suppresses.

Training and toolchain: PennyLane, Qiskit Aer, Qiskit Runtime

PennyLane: train

- PQC as a PyTorch layer (`TorchLayer`); one optimiser trains all weights.
- Adjoint/backprop gradients on a GPU statevector (16 qubits = 1 MB).

Qiskit Aer: emulate

- Ideal statevector check against PennyLane.
- Noise model from the `ibm_fez` calibration predicts the hardware gap before using QPU time.

Qiskit Runtime: run

- Transpile to native gates (CZ, RZ, SX, X).
- Estimator primitive returns the 16 $\langle Z_i \rangle$; built-in error mitigation.

Why the PQC is frozen on hardware

Parameter-shift gradients, $\partial_\phi f = \frac{1}{2}[f(\phi + \frac{\pi}{2}) - f(\phi - \frac{\pi}{2})]$, need 2 circuits per angle: $2 \times 96 = 192$ runs per sample per step. We pretrain on the simulator and fine-tune only the classical head on hardware.

Same circuit throughout: PennyLane $\rightarrow$ Aer (ideal, noisy) $\rightarrow$ IBM hardware.

Physics model: coupled flow, swirl and heat

Steady, axisymmetric, incompressible Navier–Stokes with Boussinesq buoyancy (nondimensional):

$$\begin{aligned}
\text{mass:} \quad & \partial_r(ru_r) + r\,\partial_z u_z = 0 \\
r\text{-mom.:} \quad & u_r u_{r,r} + u_z u_{r,z} - \tfrac{u_\theta^2}{r} + p_{\text{coeff}}\, p_{,r} - \tfrac{1}{Re}\left[u_{r,rr} + \tfrac{u_{r,r}}{r} - \tfrac{u_r}{r^2} + u_{r,zz}\right] = 0 \\
z\text{-mom.:} \quad & u_r u_{z,r} + u_z u_{z,z} + p_{\text{coeff}}\, p_{,z} - \tfrac{1}{Re}\left[u_{z,rr} + \tfrac{u_{z,r}}{r} + u_{z,zz}\right] - \tfrac{Gr}{Re^2}\,\theta = 0 \\
\text{swirl:} \quad & u_r u_{\theta,r} + u_z u_{\theta,z} + \tfrac{u_r u_\theta}{r} - \tfrac{1}{Re}\left[u_{\theta,rr} + \tfrac{u_{\theta,r}}{r} - \tfrac{u_\theta}{r^2} + u_{\theta,zz}\right] = 0 \\
\text{energy:} \quad & u_r\theta_{,r} + u_z\theta_{,z} - \tfrac{1}{Pr\,Re}\left[\theta_{,rr} + \tfrac{\theta_{,r}}{r} + \theta_{,zz}\right] = 0
\end{aligned}$$

$$\theta = (T - T_{\min})/(T_{\max} - T_{\min}), \quad Re = 15, \quad Pr = 28.463, \quad Gr = 8000, \quad p_{\text{coeff}} = 134{,}128$$

Two-way coupling: swirl feeds r-momentum via u_θ^2/r; temperature feeds z-momentum via buoyancy.

Physics loss and hard constraints

- Second-order derivatives via `torch.autograd.grad`.
- Five residuals balanced by weighting ($\beta = 0.95$).
- Staged schedule: data only until epoch 50, then physics weight ramped $0 \to 0.05$; gradient clipping at 1.0.
- **Hard constraints:** $u_r = r \cdot \mathrm{raw}_0$, $u_\theta = r \cdot \mathrm{raw}_2$, so both vanish exactly on the axis.

Current limitation

The PQC output is detached in the physics loss, so PDE gradients do not yet reach the quantum angles.

Fix in progress

Adjoint gradients through the residual path (PennyLane).

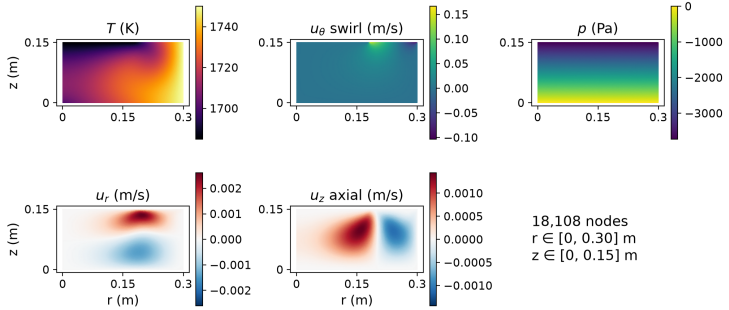
Exact constraints remove residual terms; the remaining ones are balanced, not hand-tuned.

Benchmark: silicon CZ melt

Property	Value
Reference	Axisymmetric CFD
Domain	$r \in [0, 0.30]$ m
	$z \in [0, 0.15]$ m
Nodes	18,108
Outputs	u_r, u_z, u_θ, p, T
T range	1685–1750 K
Train / val	17,203 / 905

Scope

Steady, laminar, single-phase; geometry and BCs follow Huang et al. (2025).



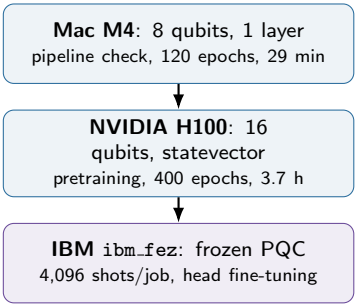
Training data: T, swirl u_θ, p, radial u_r and axial u_z. Note the counter-rotating cells and the thin swirl layer under the crystal.

Five fields spanning six orders of magnitude, from mm/s flow to kPa pressure.

Protocol and compute pipeline

- **Ablations:** classical-only at matched budget; random features in place of the PQC.
- **Baselines:** classical PINN and PINO, matched optimiser, sampling and seeds.
- **Generalisation:** hold out a whole operating condition.
- **Reporting:** per-field errors, unweighted loss terms, wall time.

Sweep	Range	Held out
T_{hot}	1730–1785 K	1780 K
ω_c (crucible)	-10 to -1 rpm	-4 rpm
ω_s (crystal)	4–20 rpm	7 rpm

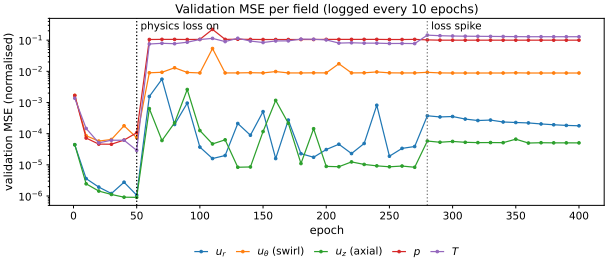


16 qubits are exactly simulable on one GPU: hardware runs test noise and deployment, not quantum advantage.

Result 1: fast convergence in the data-driven phase

Epoch	Train loss	Val MSE	Val T
1	1.16×10^{-2}	4.83×10^{-3}	1.37×10^{-3}
10	5.92×10^{-5}	3.11×10^{-4}	1.47×10^{-4}
20	5.26×10^{-5}	1.58×10^{-4}	5.07×10^{-5}
30	2.96×10^{-5}	1.75×10^{-4}	6.15×10^{-5}
50	2.07×10^{-5}	2.13×10^{-4}	2.91×10^{-5}

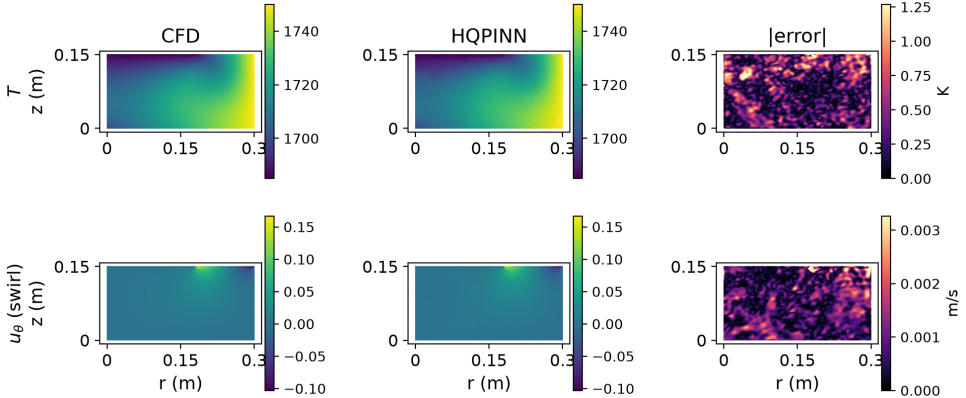
Validation MSE drops **30× in 20 epochs**; best checkpoint 1.58×10^{-4} (random spatial split).



Per-field validation MSE over 400 epochs; physics loss switches on at epoch 50.

Fast early convergence is measured; a matched classical baseline is the next test.

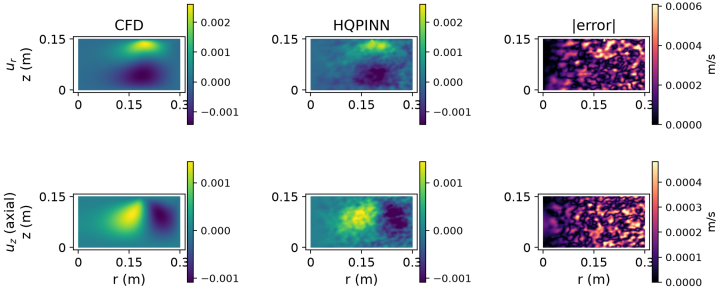
Result 2: temperature and swirl fields



CFD vs Hybrid-QPINN (best checkpoint) and absolute error: T (error $\lesssim$1.25 K on a 65 K range) and u_θ (error $\lesssim 3 \times 10^{-3}$ m/s on $\pm$0.15 m/s).

The thin swirl layer under the crystal and the hot-wall thermal gradient are both reproduced.

Result 3: velocities, fields



u_r (top) and u_z (bottom): CFD, Hybrid-QPINN, |error|.

Field	Val MSE (norm.)
u_r	1.93×10^{-6}
u_z	5.78×10^{-5}
u_θ	1.44×10^{-6}
p	4.62×10^{-5}
T	5.07×10^{-5}

Counter-rotating cells are located correctly; the remaining error is small-scale noise.

All five coupled fields are fitted in distribution at the best checkpoint.

Result 4: naive physics weighting destabilises training

Field	Best (ep. 20)	Final (ep. 400)	Factor
u_r	1.93×10^{-6}	1.79×10^{-4}	93×
u_z	5.78×10^{-5}	8.83×10^{-3}	153×
u_θ	1.44×10^{-6}	5.06×10^{-5}	35×
p	4.62×10^{-5}	0.100	**2,165**×
T	5.07×10^{-5}	0.129	**2,544**×

```
[0280] loss=1.04e+09 data=5.06e-02 phys=2.16e+10
[0290] loss=1.18e+14 data=4.97e-02 phys=2.15e+15
```

Diagnosis

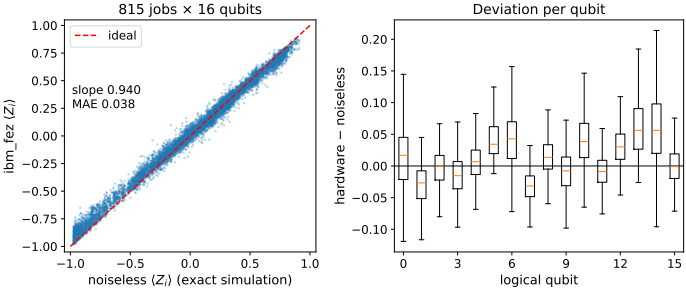
Data loss stays near 0.05; the blow-up is entirely in the physics residual, caused by the p_{coeff} scale mismatch (Bottleneck 3).

Fixes under test

Physics weight 0.001–0.005; per-field pressure scaling.

A classical-side issue, not a quantum one: the PQC is detached from the physics loss.

Result 5: the circuit on real quantum hardware



ibm_fez vs exact simulation, 815 jobs × 16 qubits: slope 0.940, MAE 0.038.

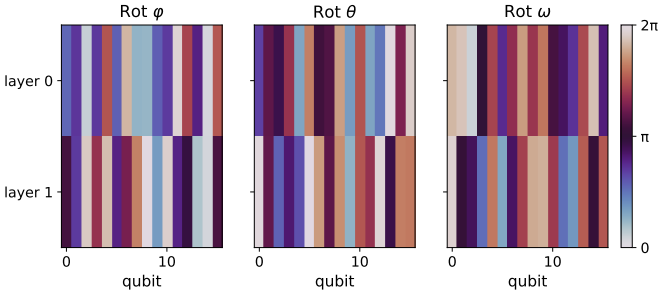
Fine-tuning session	Value
Backend	ibm_fez (156 qubits)
Jobs completed / failed	**500 / 0**
Shots per job	4,096
Session	125 min
Trainable scope	head (19,205)

Reading

Shot noise is at most 0.016, so the 0.038 gap is device noise (gates, readout), which shrinks $\langle Z \rangle$ toward 0 (slope < 1).

More shots will not close the gap; error mitigation and head re-adaptation will.

Inside the trained circuit



Trained Rot angles $(\varphi, \vartheta, \omega)$ per qubit, layers 0 and 1.

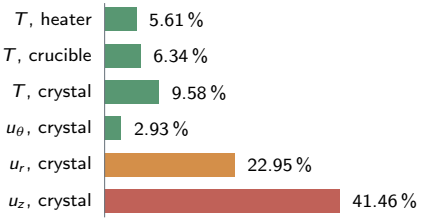
- Angles span the full $[0, 2\pi]$ range (mean 3.49, std 1.79 rad).
- No qubit stuck at a degenerate value; the two layers learn different patterns.
- LayerNorm weights qubits unequally ($\gamma \approx 0.3$–1.2) in longer runs.

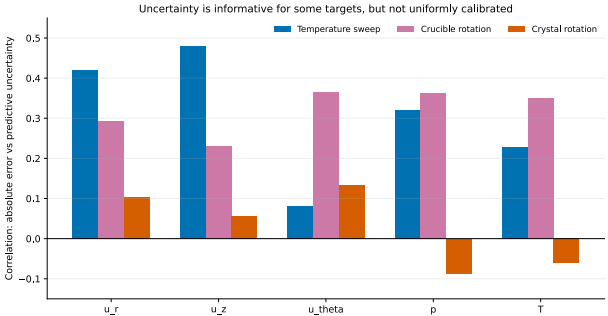
The circuit is active and differentiated; whether it adds accuracy is for the ablation to decide.

Generalisation baseline: classical surrogate on unseen conditions

Gaussian-process surrogate

Learns $(r, z, s) \mapsto$ fields; one whole case per sweep held out.
NRMSE = RMSE / field std.





Correlation between GPR predictive uncertainty and actual error, per field and sweep.

T generalises (5.6–9.6%), velocities do not (23–41%): the bar QPINO must beat.

Current status

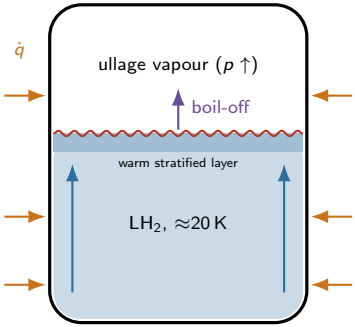
Measured so far	
Hybrid pipeline runs on GPU and real QPU	500/500 IBM jobs; MAE 0.038 vs simulation
Five coupled CZ fields fitted	Val MSE 1.58×10^{-4}, $30\times$ drop in 20 epochs
Physics loss needs field-aware scaling	$2{,}000\times$ degradation for p and T
Still to be tested	
Faster convergence than PINN/PINO	Match the classical baseline
PQC adds accuracy over classical features	Classical-only and random-feature ablations
Sharper generalisation	Case-wise holdout vs GPR / PINO (QPINO)
Reduced computational cost	GPU-hours to target error
Concentration, interfaces, LH_2	Species equation, interface tracking

Early indications are positive; matched ablations will decide the hybrid-vs-classical question.

Next benchmark: LH_2 cryogenic boil-off

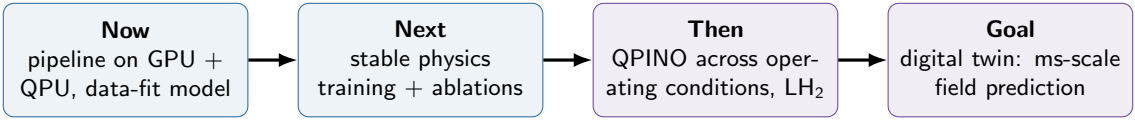
Physics to capture

- Wall heat leak $\rightarrow$ natural-convection boundary layer and thermal stratification.
- Turbulent convection (RANS in the reference CFD).
- Liquid–vapour interface with evaporation (VOF + phase change).
- Outputs: velocity, temperature, phase fraction, boil-off rate, ullage pressure.



Same framework, harder physics: turbulence plus a moving phase boundary.

Real-time digital twins



QPINO: one model for the whole process window

Process parameters become inputs: $(r, z, s) \mapsto (u_r, u_z, u_\theta, p, T)$, $s \in \{T_{\mathrm{hot}}, \omega_c, \omega_s, v_{\mathrm{pull}}\}$.
Trained once on CFD cases (generated by agentic CFD, AutoFOAM), then queried for any setting.

Inference needs no QPU: the trained 16-qubit circuit is evaluated in milliseconds on a GPU.

Key takeaways

1. **Problem:** classical PINNs struggle with spectral bias, loss imbalance and field-scale mismatch in coupled transport.
2. **Model:** Fourier features + data re-uploading + a shallow 16-qubit PQC (96 angles) as a trainable feature map, built with PennyLane and Qiskit.
3. **Results:** all five coupled CZ fields fitted (val MSE 1.58×10^{-4}); the circuit runs on IBM hardware (500/500 jobs, MAE 0.038).
4. **Next:** matched ablations for convergence, generalisation and cost; QPINO for crystal growth and LH_2 digital twins.

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Thank you

Questions?

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